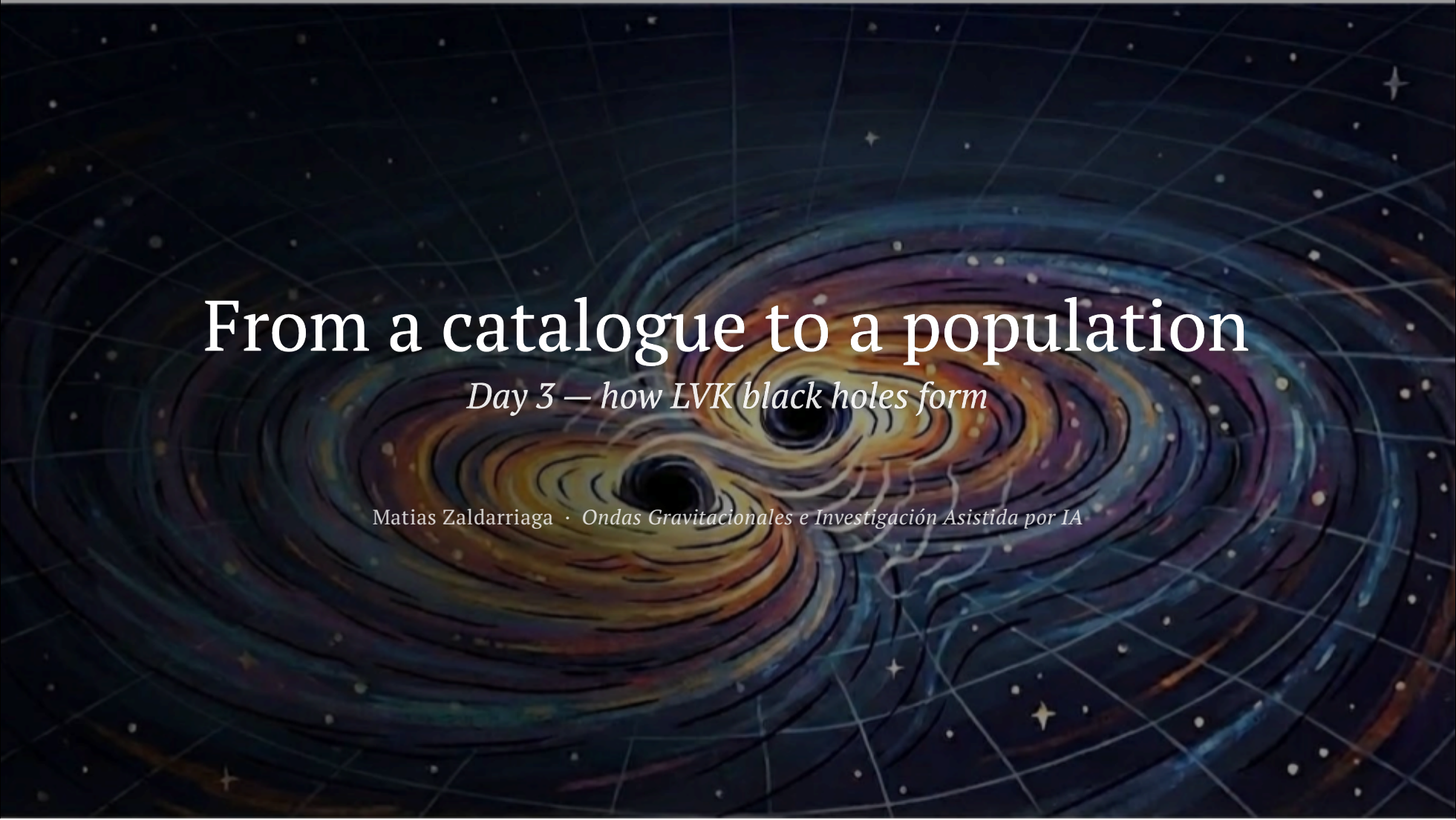




From a catalogue to a population

Day 3 — how LVK black holes form

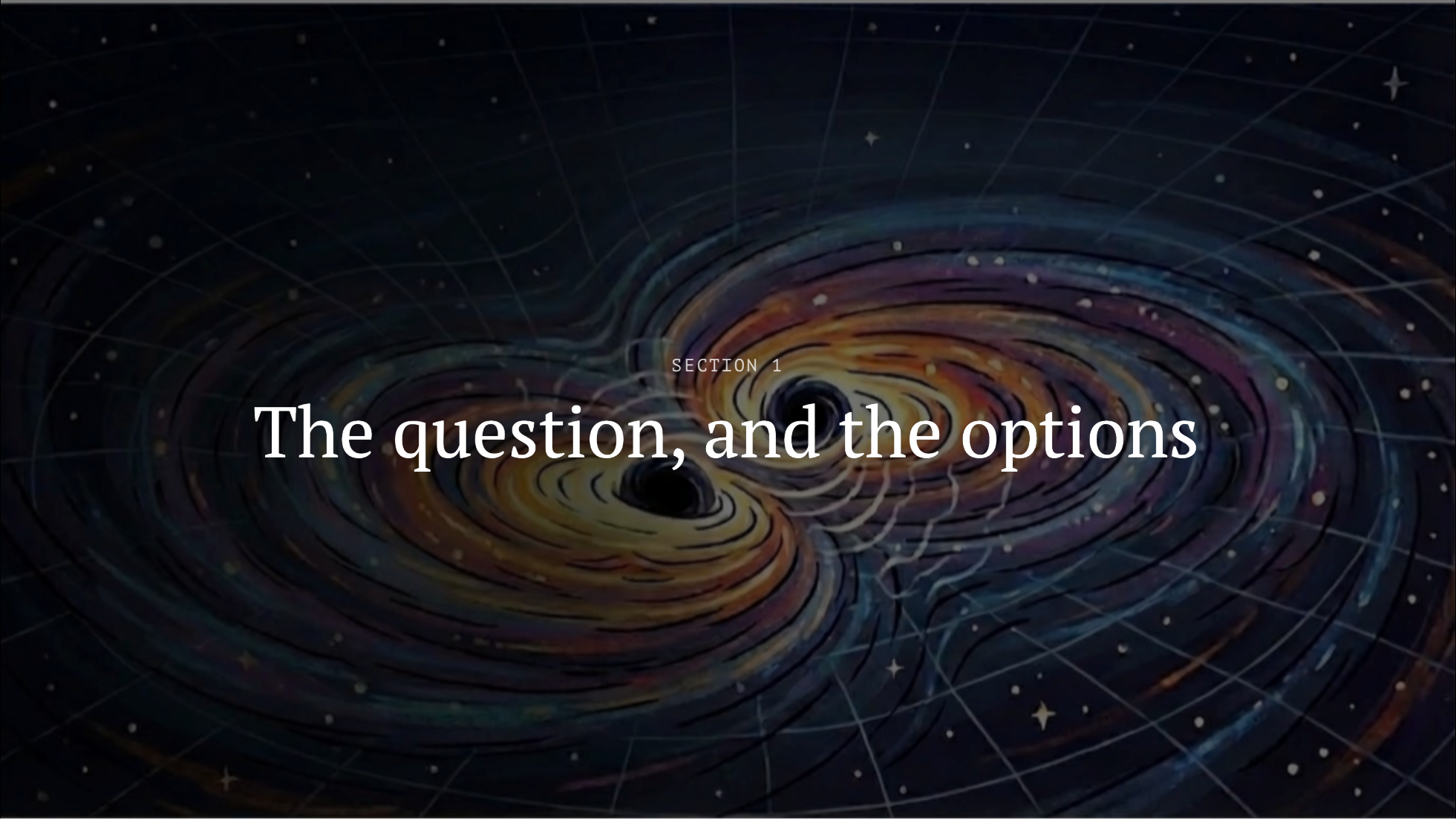
Matias Zaldarriaga · Ondas Gravitacionales e Investigación Asistida por IA

The question of the *day*.

How do the LVK binaries form? All we have is the properties of ninety-odd events, and from theory a prediction for the population. Getting from one to the other takes **three steps** — a **posterior** for each event, meaning the whole distribution its data leave over its parameters rather than one number; a selection function; and a hierarchical likelihood.

At every step the wrong answer is smooth, plausible, and comes with error bars — which is why every step today is built, not quoted.

What you can see and what is there differ by a factor of **tens** across the observed mass range. That factor is computable from first principles right up to the point where it isn't — which is why everybody runs injections.

The background of the slide is a dark, deep blue space filled with a grid of thin, light blue lines. Two large, swirling galaxies are the central focus, their colors transitioning from dark blue and purple on the outside to bright yellow and orange near their cores. The galaxies appear to be in the process of merging, with their spiral arms interacting. Numerous small, white, four-pointed stars are scattered across the dark background.

SECTION 1

The question, and the options

How do these binaries form?

ISOLATED BINARY EVOLUTION

Two massive stars, born together, stay together. Common-envelope phase, then a tight black-hole binary.

Predicts: spins roughly **aligned** with the orbital angular momentum (tides and accretion torque them into line), mass ratios near **equal**, and a spectrum that stops where pair-instability supernovae take over.

Two observables discriminate: **the spin tilts** and **the mass ratios**.

We will not settle it today. This lecture is about what *would* settle it, and about why the two observables that discriminate are exactly the two the detector distorts most.

DYNAMICAL ASSEMBLY

Black holes sink to the centre of a globular cluster or a nuclear star cluster and pair up by three-body encounters.

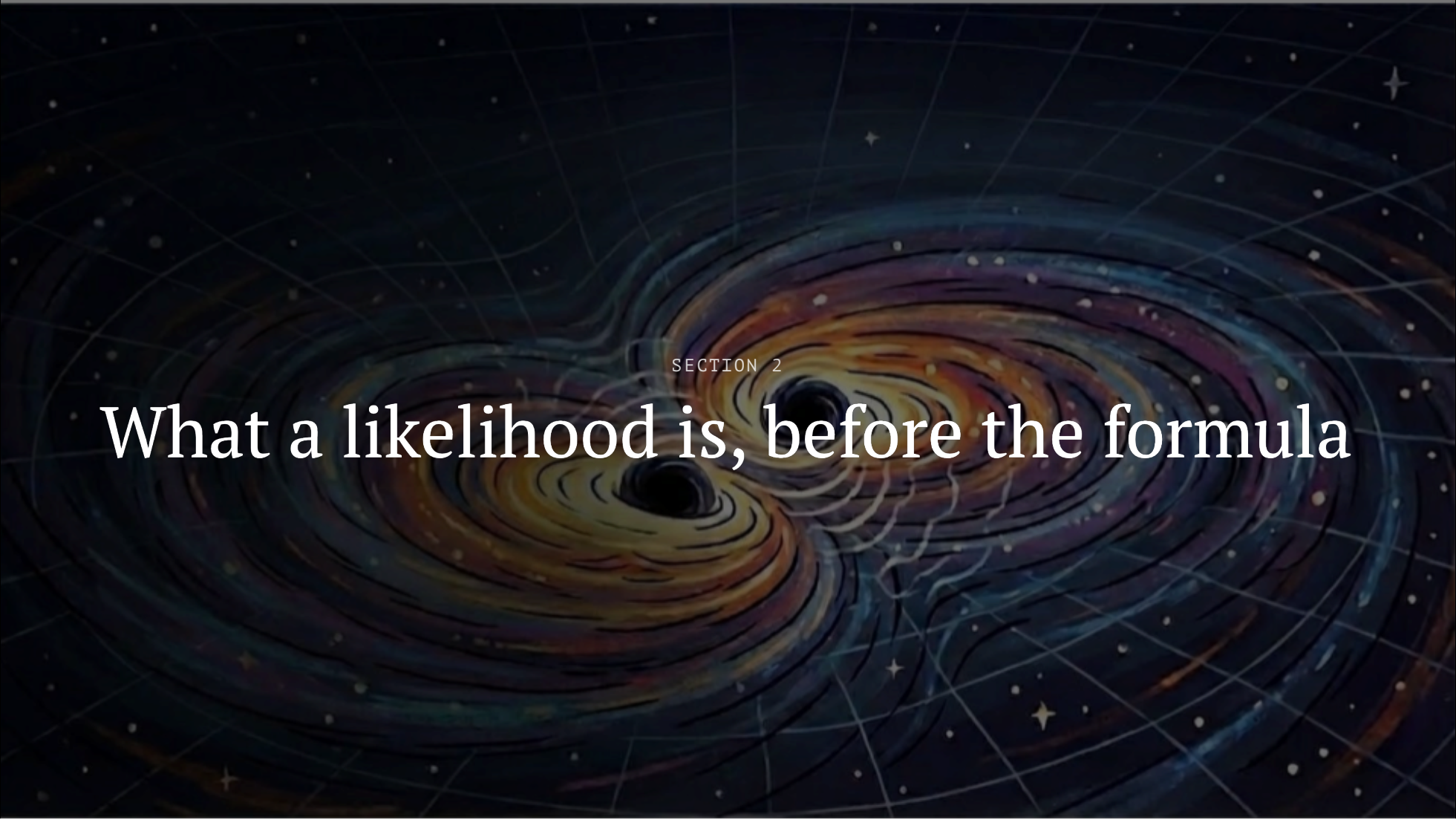
Predicts: isotropic spin tilts — the binary has no memory of how its members were born — **broad** mass ratios, and hierarchical mergers that can populate the gap.

The chain, on one slide

strain → single-event posterior → catalogue → population

arrow	what it costs	today
strain → posterior	a 15-dimensional stochastic sampler, hours per event	beats 3–5
posterior → catalogue	a detection statistic, and a threshold you have to justify	beat 7
catalogue → population	a selection function and a hierarchical likelihood	beats 6–7

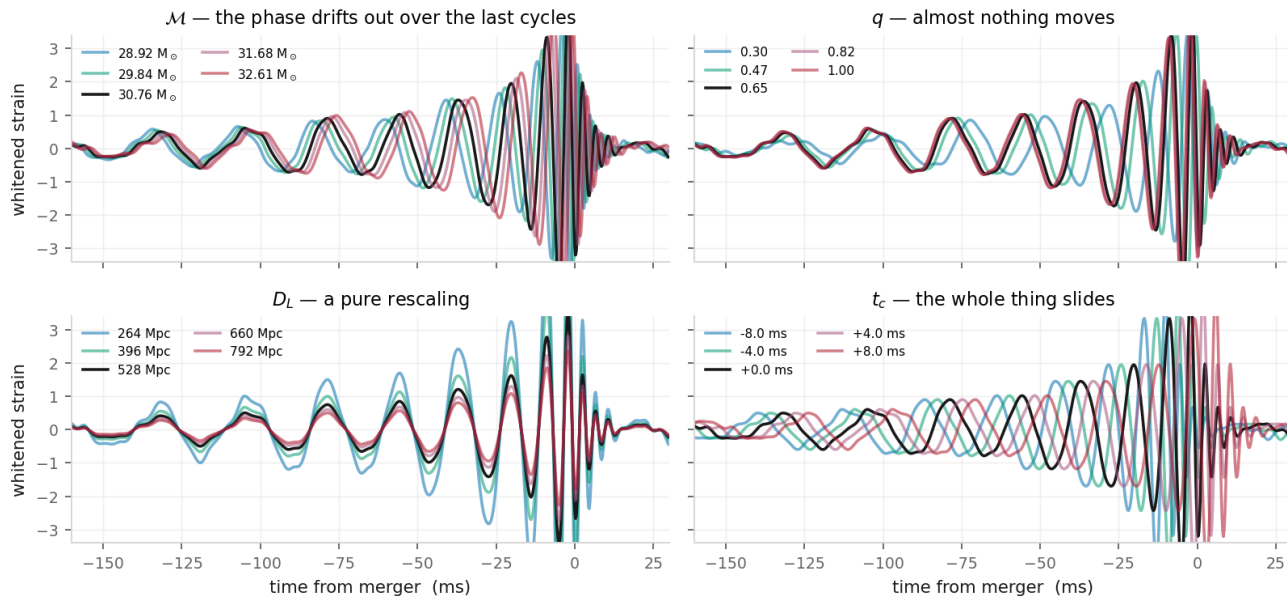
We will come back to this slide five times. Every failure mode today is an arrow somebody skipped.



SECTION 2

What a likelihood is, before the formula

Vary one parameter at a time and watch



Beat 3, before any formula. Vary one parameter at a time and watch what it does to the waveform. The chirp mass drifts the phase out over the last cycles -- a 6 per cent error costs 0.024 in match; the mass ratio barely moves anything -- halving it costs 0.050; the distance is a pure rescaling and costs EXACTLY zero, because it enters only through the amplitude. This predicts the shape of the posterior before anyone writes it down: the parameters that visibly move the waveform are the ones that come out tight. Shown whitened over the last 160 ms because that is what a person can look at; every number is computed over the full 32 s segment and the full analysis band.

Whitened templates, last 160 ms before merger, all at the same amplitude and alignment. One parameter moves per panel.

What each parameter did

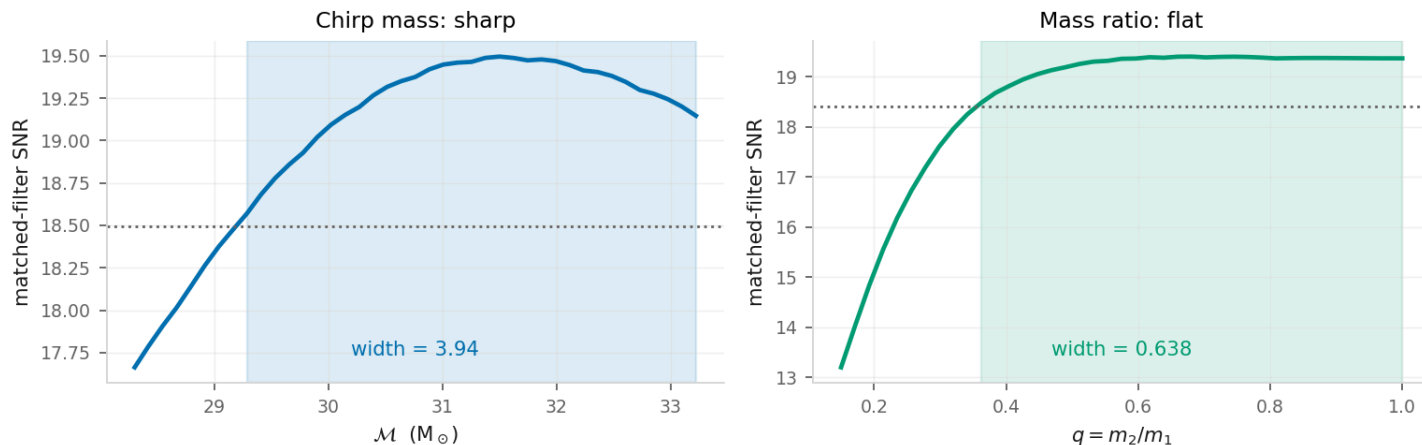
parameter	what it does to the model	loss in match
chirp mass	the phase drifts out over the last cycles	0.024 for 6 %
mass ratio	almost nothing moves	0.050 for a halving
distance	a pure rescaling — the shape is untouched	exactly zero
arrival time	the whole thing slides	—

This predicts the shape of the posterior before anyone writes it down.

The parameters that visibly destroy the fit are the ones that come out tight.

Distance enters *only* through the amplitude. Hold that thought — it is the whole of the next section.

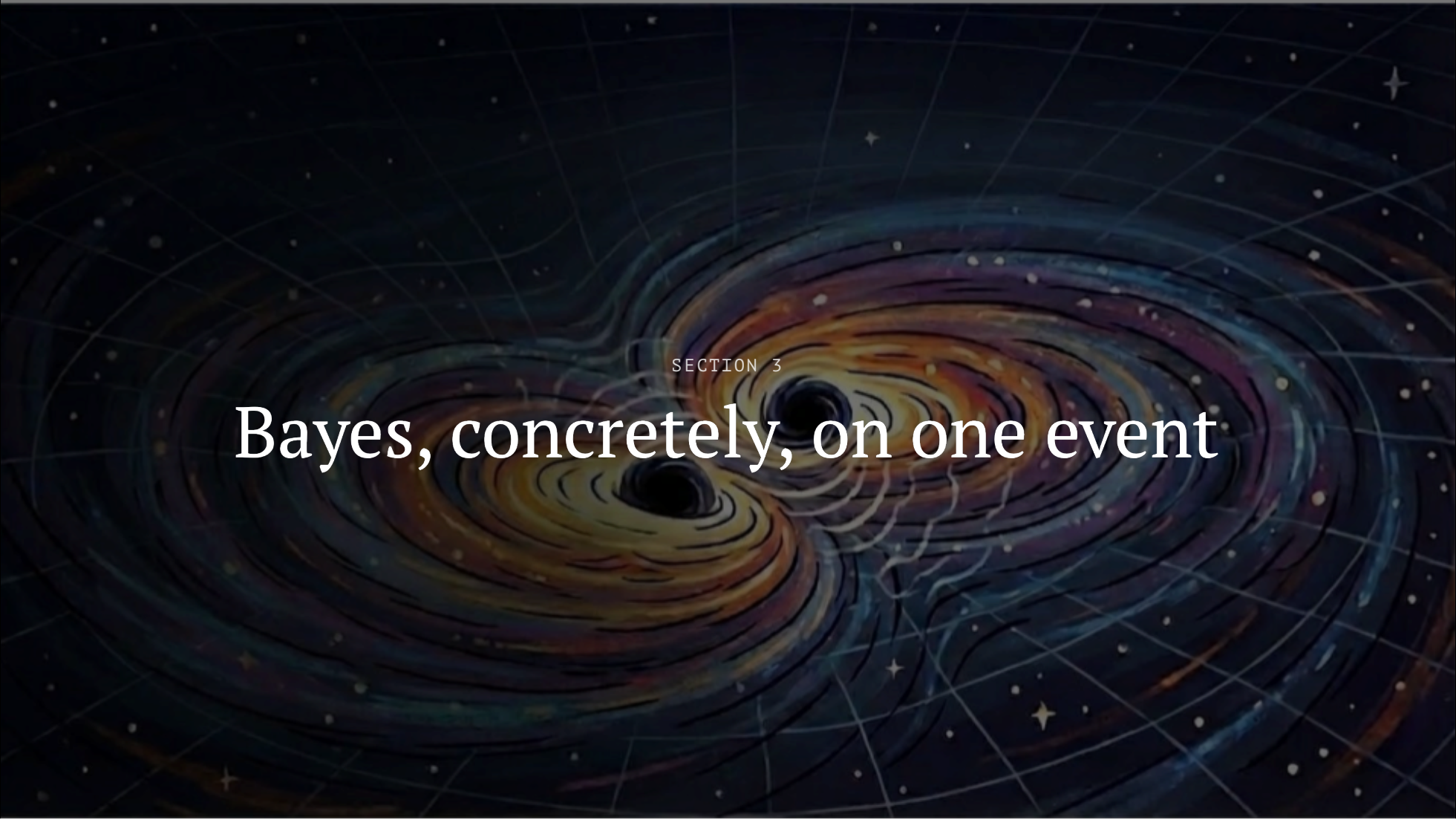
The same claim, as a number



The same claim as the sweeps, as a number. Shaded is where the SNR is within 1 of its peak. As a fraction of the range each parameter is searched over, the chirp mass is pinned to 4.1 per cent of $[4.3, 100] M_{\text{sun}}$ and the mass ratio to 75 per cent of $[0.15, 1]$ -- a factor of 18. The chirp-mass half-width, $1.97 M_{\text{sun}}$, agrees with day 2's independently measured 0.97-match half-width of $1.29 M_{\text{sun}}$ to a factor of 1.5, which is as close as two different criteria on two different quantities should be expected to come.

Shaded: where the matched-filter SNR is within 1 of its peak.

As a fraction of the range each is **searched over**: the chirp mass is pinned to **4.1 %** of $4.3\text{--}100 M_{\odot}$, the mass ratio to **75 %** of $0.15\text{--}1$. A factor of **18**.



SECTION 3

Bayes, concretely, on one event

Bayes' theorem.

You have a model with parameters θ and a stretch of data d . What you want is the distribution of θ given d . What you can write down is the distribution of d given θ — that is the likelihood, and day 2 built it. The two are related by one line of conditional probability:

$$p(\theta \mid d) = \frac{p(d \mid \theta) p(\theta)}{p(d)}$$

LIKELIHOOD

$p(d \mid \theta)$ — what the data say.
Day 2's $\ln L = -\frac{1}{2} \sum (d_i - \theta)^2$.

PRIOR

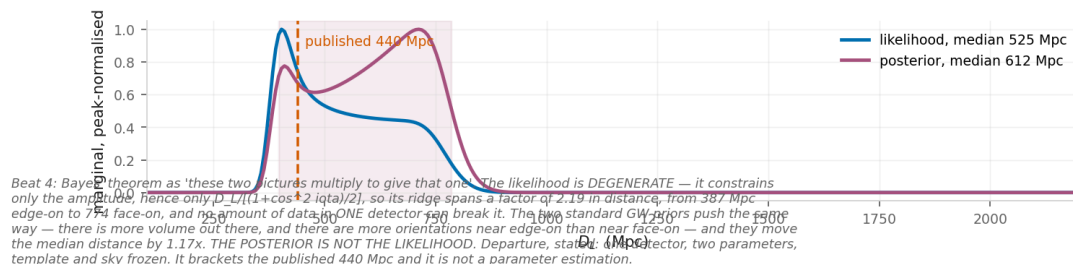
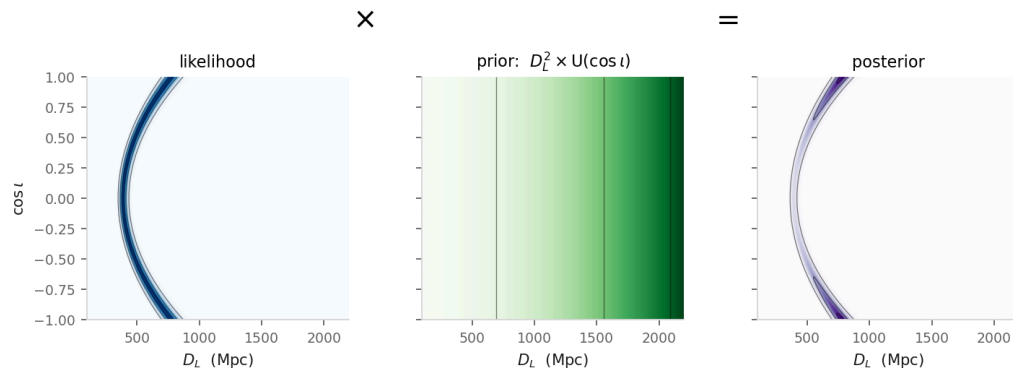
$p(\theta)$ — what you assumed before looking.

POSTERIOR

$p(\theta \mid d)$ — their product, normalised by $p(d)$, which does not depend on θ .

Everything for the rest of the day is that equation, applied twice: once per event, and then once to the whole catalogue.

Three surfaces on one *grid*.



Likelihood, prior and posterior for GW150914 on the same distance-inclination grid.

Bayes' theorem as three pictures: the **likelihood** (what the data say, day 2's $\ln L$), the **prior** (what you assumed before looking — here the volume and orientation weights), and the **posterior** — their product, normalised. These two pictures multiply to give that one.

Why the likelihood is degenerate

The data measure the **amplitude**. The amplitude is

$$a \propto \frac{1}{D_L} \cdot \frac{1 + \cos^2 \iota}{2}$$

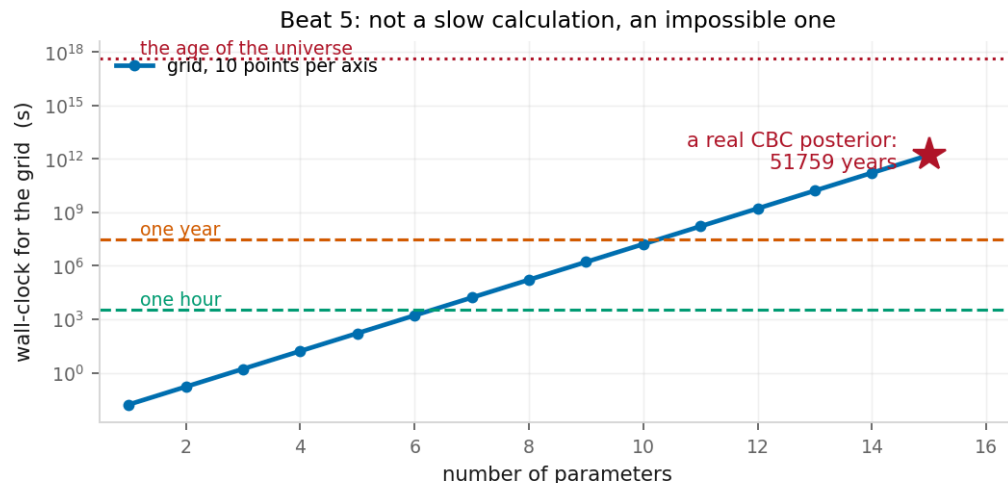
so one detector constrains that **combination** and nothing else. The ridge runs from **387 Mpc** edge-on to **774 Mpc** face-on: a factor of 2, and no amount of data breaks it.

The two priors that act on that ridge push the **same way**:

- $\pi(D_L) \propto D_L^2$ — there is more volume out there
- $\pi(\cos \iota)$ uniform — there are more orientations near edge-on than near face-on

Median distance moves **525 → 612 Mpc**. The posterior is not the likelihood.

Why samples, and not grids



A likelihood evaluation here costs 1.63 ms — MEASURED on this machine, 1.47 ms of waveform plus 0.17 ms of inner product — so ten points per axis in fifteen dimensions is 51759 years, and ten points per axis would be a terrible grid. This number is a Ledger entry and not a citation: [wiki/day3-plan.md](#) quotes 3.09 ms for a folded cogwheel evaluation on a different machine with a different likelihood, and that number may not go on a slide of ours without its own measurement. The half that matters more is not on this figure at all: you never wanted the density, you wanted marginals, and marginalising samples is ignoring a column.

The same posterior by grid and by samples.

One likelihood evaluation: **1.63 ms**, measured on the machine that made this figure. Ten points per axis in fifteen dimensions is **52,000 years** — and ten points per axis would be a terrible grid.

The half that matters more

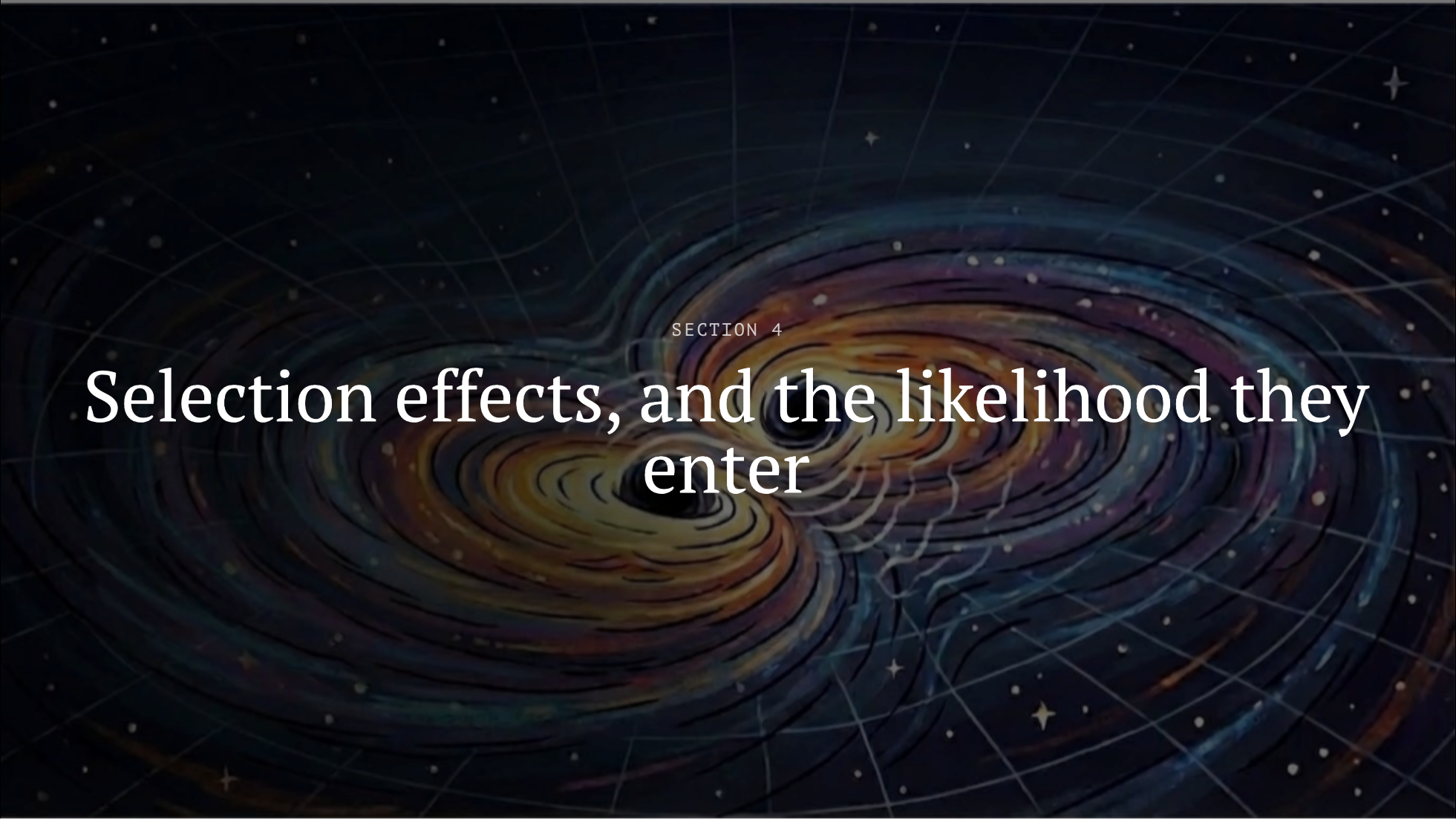
You never wanted the density.

You wanted marginals and expectations — and samples give you those by counting, at a cost that does not care about dimension.

Marginalising a grid is an integral.

Marginalising samples is ignoring a column.

The triptych two slides ago marginalised a 320×320 grid. It is the last time in this course that doing it that way is affordable.



SECTION 4

Selection effects, and the likelihood they enter

The scaling you get for free

From day 2's SNR, $\rho \propto \mathcal{M}^{5/6}/D_L$. So the horizon distance goes as $\mathcal{M}^{5/6}$ and the Euclidean sensitive volume as

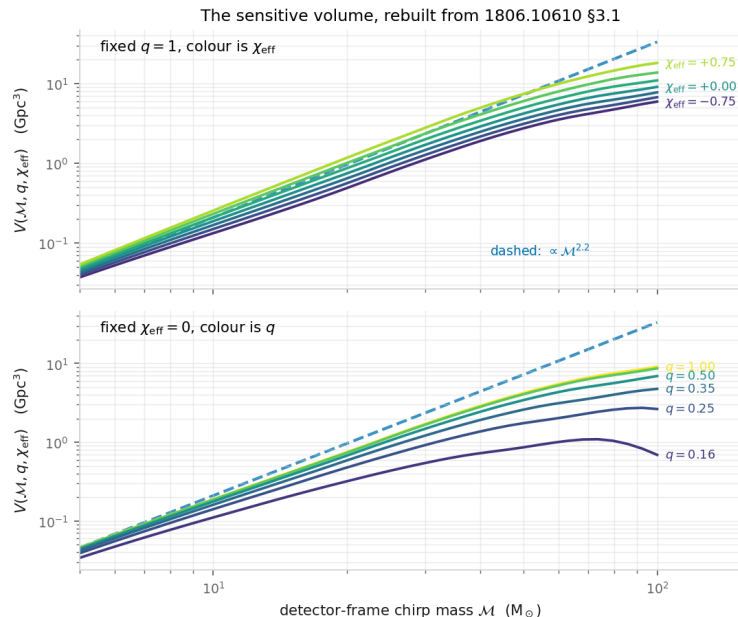
$$V \propto \mathcal{M}^{5/2}.$$

Ten minutes, no code, and it reuses machinery you built on day 2. The exponent comes from the amplitude alone — it is $5/6$ for **any** noise curve, as long as the band is held fixed.

The measured answer is **not** 2.5. Fig. 3 of Roulet & Zaldarriaga (2019) draws a reference power law at $\mathcal{M}^{2.2}$.

Over the observed range 8–40 $M_\odot$, $\mathcal{M}^{2.2}$ is a factor of **34**. Nobody can call that a correction.

The two other axes, which are the sharper argument



Gate G1: $V(30, 1, 0) = 1.70 \text{ Gpc}^3$ against 1.86 read off the published figure, ratio 0.915. G3: $V(+0.5)/V(0) = 1.38$ and $V(-0.5)/V(0) = 0.75$. G4: flat above $q \sim 0.5$ (0.87) and collapsed below (0.32 at $q = 0.16$). You are preferentially detecting the aligned-spin systems and preferentially missing the unequal-mass ones – which are exactly the two observables that discriminate formation channel. The dashed line is the paper's reference slope; it is drawn, not fitted, and it is NOT in the gate.

Sensitive volume along the spin and mass-ratio axes.

Put those together

You are preferentially detecting the **aligned-spin** systems, and preferentially missing the **unequal-mass** ones.

At $\mathcal{M} = 30 M_{\odot}$: $V(\chi_{\text{eff}}=+0.5)$ is $1.38\times V(0)$, and $V(q=0.16)$ is $0.32\times V(q=1)$.

Those are **exactly the two observables that discriminate formation channel**.

Selection pushes both of them toward the field-binary answer. That is a much stronger statement than "selection tilts the mass function": the bias sits directly on the quantity we are trying to measure, not off to one side in a nuisance parameter.

Then the retreat — and the paper does it in its own voice

Whether something enters a catalogue depends on glitch rejection, detector artifacts, PSD drift, **how much your signal resembles the glitches this particular detector happens to make**, the fact that the catalogue is built from multiple detectors so the answer depends on both PSDs and on sky location, and on duty cycle.

Roulet & Zaldarriaga §2 concedes all of it while doing the calculation:

- the reference PSD is *"the harmonic mean of the combined-channel PSDs of the first six events"* — one fixed curve for two detectors over two runs
- detection is $\rho > 9$ on the **expectation value**, with noise fluctuations *"important only near the boundary of the sensitive volume, and we ignore it for simplicity"*
- and a closing paragraph conceding that template-bank effectualness varies too

Conclusion: you do injections.

The population likelihood, *built*.

Take every merger in the universe to be an independent draw — a Poisson process whose rate varies with the parameters. A population model Λ says how many and where, and the sensitive volume from ten minutes ago says which of them you see:

$$\frac{dN}{d\theta} = R T p(\theta \mid \Lambda), \quad N_{\text{exp}}(\Lambda, R) = R T \int d\theta p(\theta \mid \Lambda) P_{\text{det}}(\theta) \equiv R \langle VT \rangle(\Lambda)$$

A Poisson process with N points contributes $e^{-N_{\text{exp}}}$ times one factor per point. You never see θ_i , only d_i , so each factor is **day 2's likelihood** integrated against the model:

$$\mathcal{L}(\{d\} \mid \Lambda, R) = e^{-R \langle VT \rangle(\Lambda)} \prod_{i=1}^N R T \int d\theta p(d_i \mid \theta) p(\theta \mid \Lambda)$$

Marginalise the overall rate R under a log-uniform prior. That integral is a Gamma function, the R^N and the exponential cancel it, and the selection function comes back downstairs, once per event:

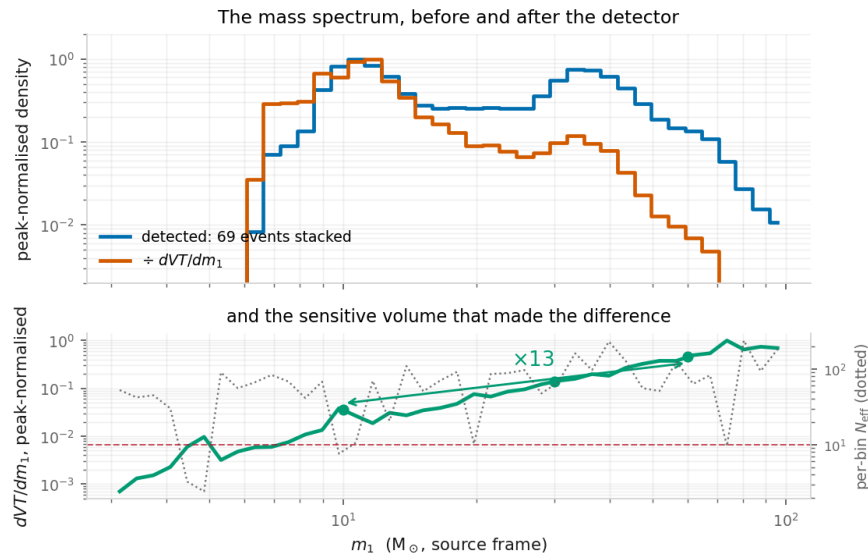
$$\mathcal{L}(\{d\} \mid \Lambda) \propto \prod_{i=1}^N \frac{\int d\theta p(d_i \mid \theta) p(\theta \mid \Lambda)}{\langle VT \rangle(\Lambda)}$$



SECTION 5

What the population looks like

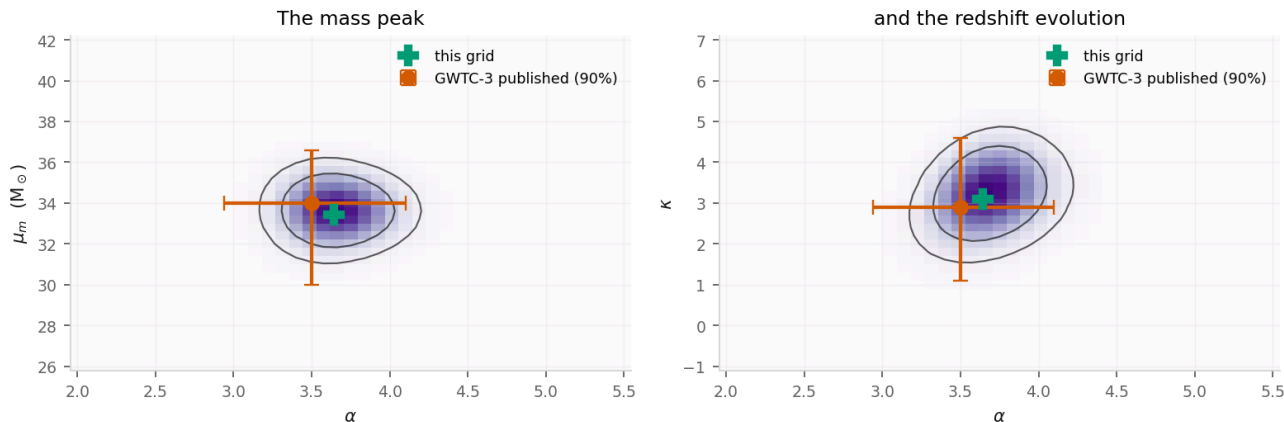
What you see, and what the detector let you see



The sensitive volume-time per unit primary mass rises by a factor of 3.9 between $m_1 = 10$ and $30 M_\text{sun}$ and 12.6 between 10 and 60, measured from the LVK found-injection set. So the observed mass distribution is tilted by more than an order of magnitude before any astrophysics enters -- nobody can call that a correction. Note what the curve is NOT: the found FRACTION among the injections in the file falls with mass, because the file holds only the accepted injections and acceptance already favours loud systems. Dividing by dVT/dm_1 is the crude way to undo the tilt -- what the hierarchical likelihood reduces to if every event were measured perfectly -- and the dotted line is why it is crude: the per-bin N_{eff} falls to 2.4 at $m_1 = 4.9 M_\text{sun}$, and dividing by a number estimated from that many effective injections is how a histogram grows a feature that is not there.

The observed masses against the selection-weighted spectrum.

Done properly

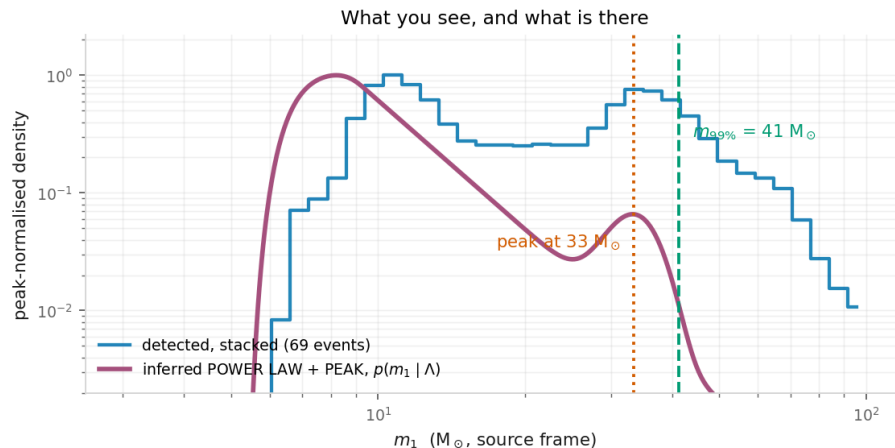


POWER LAW + PEAK fitted to 69 events with the selection integral in the denominator, on two 2-D slices with the other seven hyper-parameters frozen at the shipped medians. $\alpha = 3.64$, $\mu_m = 33.4 M_\odot$, $\kappa = 3.11$, all inside the published 90 per cent intervals. THAT AGREEMENT IS THE TEST OF SOMETHING ELSE: the shipped 'weights' field could not be identified as $1/\pi$ PE (it regresses to a pure $(1+z)^{-4.105}$ with no volume factor), so the PE prior was reconstructed from the collaboration's stated one, and this figure is the only thing standing behind that choice. Our intervals are 0.70x the published width because frozen parameters cannot absorb anything -- a property of the method, not a better measurement.

POWER LAW + PEAK refit against the published intervals.

POWER LAW + PEAK, 69 confident BBH, with the selection integral in the **denominator of the likelihood** rather than as a reweighting. Everything lands inside the published 90 % intervals.

The spectrum and the rate

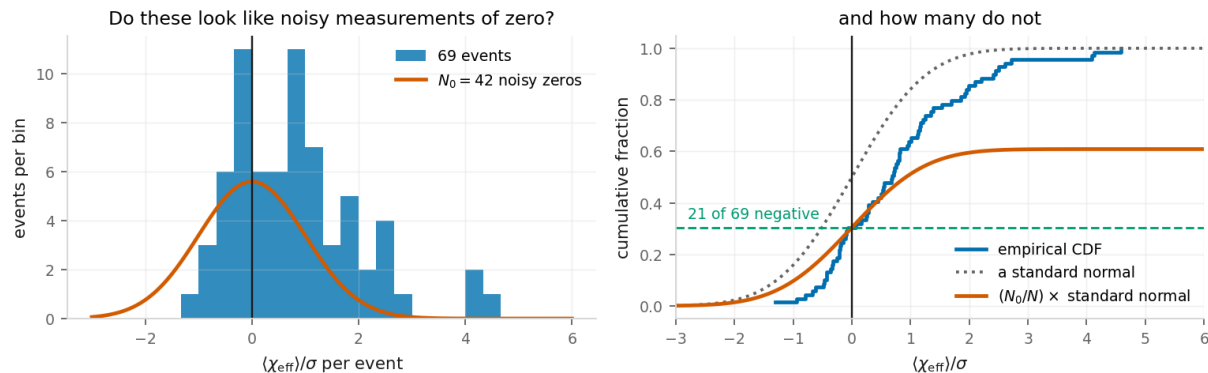


The inferred spectrum falls far more steeply than the detected one, because the detector's reach rises with mass. $m_{99\%}$ -- the mass below which 99 per cent of primaries lie -- comes out at $41 M_\odot$ against the published $44.5.1/+9.2$, down from GWTC-2's 60. Note what $m_{99\%}$ is NOT: it is a property of the fitted distribution, not the m max cutoff parameter ($88 M_\odot$ here), which the data barely constrain. Confusing the two is how 'no gap above 60' gets misread as 'the spectrum stops at 44'.

The inferred mass spectrum and rate evolution.

$\alpha = 3.5$, a Gaussian peak at $34 M_\odot$, $m_{99\%} = 44 M_\odot$ — down from 60 in GWTC-2 — **no gap above 60**, chirp-mass over-densities at 8.3 and 27.9, and $\kappa = 2.9^{+1.7}_{-1.8}$ for the redshift evolution.

Spins — before any model at all



The paper's headline claim, and it needs NO population machinery and no selection correction: are the per-event measurements noisy measurements of zero? 21 of 69 sit below zero, so about 42 events (61 per cent) are consistent with being noisy zeros and 27 are in excess on the POSITIVE side. Against the paper's ~20 of 30 (67 per cent) with ~10 in excess and none requiring a negative tail. The fraction reproduces; the counts are not expected to, because their gold sample is 30 of 55 selected by FAR and by the absence of a nearby non-Gaussian transient and this is all 69 confident BBH. Note how N_0 is obtained: if the noisy zeros are symmetric about 0 and the excess is all positive, $N_0 = 2 \times$ (the number negative). A fitted version of this returned 'all of them', because it was measuring its own model.

Per-event effective-spin intervals, no model assumed.

No population machinery, no selection correction. **21 of 69** events sit below zero, so about **42** are consistent with being noisy measurements of $\chi_{\text{eff}} = 0$ and **27 are in excess on the positive side** — and none requires a negative tail.

Counting is where the agreement stops. Put a population model on the same events and what fraction sits below zero becomes a statement about **the model you chose**, not about the events — two published analyses agree on the width of the distribution and disagree on what it means. That argument is written out in the wiki rather than adjudicated here.

Beyond this machinery's *reach*.

- **There is no eccentricity population constraint** anywhere in this group's output. The eccentricity work is waveform and surrogate modelling; nobody constrains an eccentric fraction. So eccentricity is not a third vote on formation channel — it is at most "here is the machinery that would let you ask".
- **There is no population-synthesis paper in the set.** The channel predictions on the options slide are borrowed, each with one line saying where it came from.

Saying which questions the machinery **cannot** answer is part of knowing what it does.

Where this leaves us

We did not settle how these binaries form. We built the machinery that could, and found that:

- the detector distorts **exactly** the two observables that discriminate
- the distortion is computable from first principles right up to the point where it isn't
- and past that point the honest move is an injection campaign, which is why everybody runs one